Meng Gao

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Education

Columbia University — New York, NY
Master of Science in Computer Science; GPA: 3.76/4.00 — *Sep 2024 – Dec 2025*

The Ohio State University — Columbus, OH
Bachelor of Science in Computer Science and Engineering; GPA: 3.77/4.00 — *Aug 2021 – Aug 2024*

Experience

Microsoft — United States
Software Engineer, Azure Core — *Apr 2026 – Present*

- Drove the Rust client SDK to parity with the C# and Python SDKs (TLS with certificate validation, atomic release commits, concurrent streaming downloads, lease-based disk cleanup), growing its tests from 21 to ~400.
- Extended the SDK to C/C++ services via a C ABI with restart recovery to the last accepted release and a build-sign-publish NuGet pipeline; made the E2E harness SDK-pluggable so one suite validates all four SDKs.
- Remediated 6 security-review findings across the Rust agent and C#/Python/Rust SDKs: TLS certificate validation (SAN matching, root pinning), path-traversal guards, loopback-only defaults, and SHA-256-verified staging.
- Cut end-to-end CI time from ~4 hours to ~1 hour by isolating shared test state so suites run 5-way parallel, and built change-scoped PR pipelines that validate SDK-only changes in ~44 min vs. ~105 min for a full build.
- Led Linux readiness for the agent and Python SDK, building a language-agnostic .NET launcher (JSON over stdio) so Python and Rust tests reuse ~600 lines of C# agent-lifecycle logic; enabled Linux tests in official CI (57/57 passing).
- Automated agent delivery with a pipeline that packages signed binaries into Windows/Linux Docker images, publishes to dev and approval-gated prod registries, and deploys daily to an AKS test cluster via Helm, replacing manual pushes.
- Root-caused concurrency bugs across the agent and SDKs, including a stale-signal race that tore down freshly reconnected gRPC sessions and a connection-dedup bug behind a 14.5% flaky-test rate.

Amazon Web Services — Arlington, VA
Software Development Engineer Intern — *Jun 2025 – Sep 2025*

- Designed a Redis cache-aside layer for AWS Lambda and Glue workloads, cutting external API calls by over 90%.
- Built URL-state persistence and advanced filters, lifting link sharing by 40% and positive user feedback by 20%.
- Moved hard-coded configs to AWS AppConfig feature flags, enabling controlled rollouts and instant rollback.
- Wrote 40+ integration tests that caught edge cases pre-deployment, saving ~3 hours of manual QA per release.

Columbia Build Lab — New York, NY
Full Stack Developer Intern — *Feb 2025 – May 2025*

- Built a React and Flask MVP for an early-stage startup, enabling investor demos and customer-feedback cycles.
- Designed a high-concurrency WebSocket ingestion layer for low-latency device-monitoring data.
- Set up GitHub Actions CI/CD with unit and Cypress E2E tests, cutting change-induced production failures by 90%.

Projects

Speech Emotion Recognition Pipeline | *Python, scikit-learn, Parselmouth, openSMILE* — Sep 2024 – Dec 2024

- Built an end-to-end pipeline classifying 7 speech emotions from pitch, intensity, and IS09 acoustic features.
- Benchmarked SVM, Random Forest, and MLP; a tuned RBF-kernel SVM achieved the best cross-validation accuracy.

Technical Skills

Languages: Rust, Python, C#, C/C++, Java, SQL, JavaScript, Bash, PowerShell
Cloud & DevOps: Azure (AKS, ACR, Key Vault, Pipelines), AWS (Lambda, Glue), Docker, Kubernetes, Helm, Linux
Systems & Testing: gRPC, Protobuf, TLS/X.509, C ABI/FFI, Tokio, pytest, NUnit, Cypress, Distributed Systems
Frameworks & Data: .NET, React, Flask, PyTorch, scikit-learn, Redis, Elasticsearch, MySQL, MongoDB